Mini-Case 1 | THE SWISS WATCH INDUSTRY

[9]Swiss watches are legendary. Swiss watches once dominated the world: in the mid 1970s, 43% of the units sold worldwide were Swiss made. Then things changed. By 1983, Swiss market shares plunged to less than 15%, and 1000 of the 1600 Swiss watch companies closed. These massive closures were accompanied by a contraction of the labor market by two thirds.

At the brink of bankruptcy, the creditor banks of the two flagships of the Swiss watch industry, Swiss-French SSIH and Swiss-German ASU AG, asked the consulting firm Hayek Engineering for advice: *Should we shut down or can you design a new operations strategy?*

The worldwide watch industry

During the late 1970s, manufacturers from Japan and Hong Kong penetrated the watch market by using low-cost labor, quartz technology (developed, but not used in Switzerland) and mass production. Their inexpensive watches quickly gained an important market share. To compensate for the lost revenues from diminished market share, Swiss watchmakers raised prices.

By 1980, the world market for watches was about 500 million units per year and consisted of three segments. The low-end segment had prices up to about $75 and represented 450 million units out of 500 million. The middle segment with prices up to $450 represented 42 million units, leaving 8 million watches for the top segment.

Margin retreat left the Swiss with zero share of the low-end segment; only 3% of the middle segment; but 97% share of the luxury watches (which accounted for 0.4% of world unit watch sales).

Watch technology

Mechanical watches consist of two groups of parts: the visible exterior parts such as the case, the dial, and hands, as well as the interior "movement," which contains about 200 individual parts and is driven merely by a spring. After a long evolution, two main mechanical watch technologies emerged. The *lever* was used for expensive watches where jewels tipped the teeth in the movement. The *pin pallet* was used for cheaper watches.

The process to produce and assemble mechanical watches was labor intensive. Master artists would work in a library-style environment on well-lit work desks with microscopes where they would compile a piece of "living art" worth more than the sum of its parts. In the mid 20th century, most watches sold in jewelry shops for hundreds of dollars. Their full production cost consisted of approximately 60% labor, 20% materials and 20% depreciation.

In 1957, the Hamilton Watch Co of Lancaster, Pennsylvania, produced the world's first electric watch. Its traditional balance-wheel mechanism used a battery instead of a spring as power source. In 1960, the next advance in electric watches came from Bulova, a watch company founded in 1875 in New York City by Joseph Bulova, a 23-year-old immigrant from Bohemia. The Bulova corporation replaced the balance-wheel mechanism with the electronic vibrations of a tuning fork. It raised the beat of the watch from 2.5 beats per minute for a mechanical watch to nearly 2.5 million per second for an electronic watch. The tuning fork was kept vibrating by a battery-powered electronic circuit. The Bulova electronic watch was one of the first consumer products to use a transistor.

[9]Professor Van Mieghem prepared this case as the basis for class discussion rather than to illustrate either the effective or ineffective handling of a managerial situation. This case is based on an earlier version written by Professor Yehuda Bassok. Taylor (1993) and BBC (2005) are also acknowledged. No part of this case study may be reproduced without permission; direct all inquiries to permissions@vanmieghem.us

The Japanese watch industry

The electronic watch technology was embraced and enhanced by the Japanese watch industry. Since its inception in 1880, the Japanese watch industry was concentrated in the hands of three competitors: K. Hattori (which marketed the Seiko brand), Citizen and Orient. The three companies accounted for 50%, 30% and 20% of production in 1950, respectively. Japanese manufacturers used skilled labor commanding wages at a level of approximately 1/10 of those in Switzerland.

During the 1950s, Seiko pursued the goal of closing the technological gap between the Swiss products and its own products. They also developed a policy of standardizing the movements of the different watches and invested heavily in automatic equipment. This permitted Seiko to reduce labor to 35% of the full production cost of a watch in the late 1960s. Vertical integration of component manufacturing also permitted more rapid deployment of new technology and watch design. One industry observer estimated that the Japanese enjoyed a 15% to 45% variable cost advantage over the Swiss in the mid 1960s. During the late 1960s they initiated component fabrication and assembly operations abroad. Plants were located in Hong Kong, Singapore and Malaysia. Workers in all three locations earned about one third as much as did Japanese workers. In the early 1970s, Hattori-Seiko entered the American and European markets.

Meanwhile, the Swiss research lab CEH (Centre Electronique Horloger) developed a more accurate oscillator and produced the first wristwatch quartz movement in 1967. In contrast to the Swiss manufacturers who had refined their mechanical technology to catch up with Bulova, Seiko made a commitment to develop and produce both analog and digital Quartz watches. The popularity of quartz watches began to grow around 1976. By 1977, Seiko became the largest watch company in the world.

1. *How would you contrast the operations strategy of the traditional Swiss watch company with that of Seiko?*
2. *What information would you gather and what advice would you give? Would you close shop or recommend a new strategy for the Swiss watch industry? For the latter, use the framework describing your recommended value proposition and required changes in competencies and the operational system.*